UNIVERSITY NETWORK DESIGN FINAL REPORT

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UNIVERSITY NETWORK DESIGN FINAL REPORT

1. Introduction

This report presents a complete network design for a multi-campus university consisting of **six campuses**, interconnected through a **central core network**, providing secure, scalable, and high-performance connectivity. The design uses:

- **Private IP Block:** 10.0.0.0/8
- **Interior Routing Protocol:** OSPF Multi-Area
- **Exterior Routing Protocol (between campuses):** EIGRP
- **Layer-2 Domain:** VLAN-based segmentation
- **Wireless Management:** Centralized WLC + Lightweight Access Points
- **High Availability:** Redundant links, dual routers, loop-free Layer-2

The objective is to create a centralized yet scalable network that supports thousands of students, staff, servers, and wireless users across all campuses.

2. Campus Overview

Campus	Population	Subnet Assigned	OSPF Area	AP Count	Notes
Campus A	1000	10.1.0.0/16	Area 1	50	Main campus, WLC hosted here
Campus B	600	10.2.0.0/16	Area 2	30	Faculty/Library
Campus C	400	10.3.0.0/16	Area 3	20	CS & Engineering
Campus D	300	10.4.0.0/16	Area 4	20	Admin Campus
Campus E	200	10.5.0.0/16	Area 5	15	Research Dept
Campus F	150	10.6.0.0/16	Area 6	10	Remote campus

3. Addressing Plan

3.1 Primary IP Block

10.0.0.0/8

3.2 Allocation per Campus

Each campus receives a **/16**, allowing up to 65,536 addresses.

Campus	Subnet	Range
A	10.1.0.0/16	10.1.0.1 – 10.1.255.254
B	10.2.0.0/16	10.2.0.1 – 10.2.255.254
C	10.3.0.0/16	10.3.0.1 – 10.3.255.254
D	10.4.0.0/16	10.4.0.1 – 10.4.255.254
E	10.5.0.0/16	10.5.0.1 – 10.5.255.254
F	10.6.0.0/16	10.6.0.1 – 10.6.255.254

4. VLAN & Departmental Structure

Common VLANs Across All Campuses

VLAN ID	Name	Purpose
10	Admin	Staff offices
20	Faculty	Teachers
30	Student	Labs, classrooms
40	Servers	Datacenter machines
50	Wifi	AP management

VLAN Creation

```
enable
configure terminal
vlan 10
 name Admin
exit
vlan 20
 name Faculty
exit
vlan 30
 name Student
exit
vlan 40
 name Server
exit
vlan 50
 name Wifi
exit
end
```

write memory

5. Subnetting

Campus supernet allocation :16 per campus allocation

11111111.11111111.00000000.00000000

Subnet Mask:255.255.0.0/16

Octet:2nd

Subnet Generator:2^0=1

Campus A (Head Office / Data Center): 10.1.0.0/16

Campus B (Northern): 10.2.0.0/16

Campus C (Southern): 10.3.0.0/16

Campus D (Eastern): 10.4.0.0/16

Campus E (Western): 10.5.0.0/16

Campus F (Central): 10.6.0.0/16

Campus A:

10.2.0.0/16

Department	Hosts	Subnet Mask	CIDR	Network Address	Usable IP Range	Broadcast	Gateway
Students	900	255.255.252.0	/22	10.1.0.0	10.1.0.1 – 10.1.3.254	10.1.3.255	10.1.0.1
Faculty	100	255.255.255.128	/25	10.1.4.0	10.1.4.1 – 10.1.4.126	10.1.4.127	10.1.4.1
Admin	40	255.255.255.192	/26	10.1.4.128	10.1.4.129 – 10.1.4.190	10.1.4.191	10.1.4.129
Servers	4	255.255.255.248	/29	10.1.4.192	10.1.4.193 – 10.1.4.198	10.1.4.199	10.1.4.193
Wi-Fi	20	255.255.255.224	/27	10.1.4.224	10.1.4.225 – 10.1.4.254	10.1.4.255	10.1.4.225

Campus B:

10.2.0.0/16

Department	Hosts	Subnet Mask	CIDR	Network Address	Usable IP Range	Broadcast	Gateway
Students	900	255.255.252.0	/22	10.2.0.0	10.2.0.1 – 10.2.3.254	10.2.3.255	10.2.0.1
Faculty	80	255.255.255.128	/25	10.2.4.0	10.2.4.1 – 10.2.4.126	10.2.4.127	10.2.4.1
Admin	30	255.255.255.192	/26	10.2.4.128	10.2.4.129 – 10.2.4.190	10.2.4.191	10.2.4.129
Servers	4	255.255.255.248	/29	10.2.4.192	10.2.4.193 – 10.2.4.198	10.2.4.199	10.2.4.193
Wi-Fi	20	255.255.255.224	/27	10.2.4.224	10.2.4.225 – 10.2.4.254	10.2.4.255	10.2.4.225

Campus C:

10.3.0.0/16

Department	Hosts	Subnet Mask	CIDR	Network Address	Usable IP Range	Broadcast	Gateway
Students	800	255.255.252.0	/22	10.3.0.0	10.3.0.1 – 10.3.3.254	10.3.3.255	10.3.0.1
Faculty	70	255.255.255.128	/25	10.3.4.0	10.3.4.1 – 10.3.4.126	10.3.4.127	10.3.4.1
Admin	30	255.255.255.192	/26	10.3.4.128	10.3.4.129 – 10.3.4.190	10.3.4.191	10.3.4.129
Servers	4	255.255.255.248	/29	10.3.4.192	10.3.4.193 – 10.3.4.198	10.3.4.199	10.3.4.193
Wi-Fi	20	255.255.255.224	/27	10.3.4.224	10.3.4.225 – 10.3.4.254	10.3.4.255	10.3.4.225

Campus D:

10.4.0.0/16

Department	Hosts	Subnet Mask	CIDR	Network Address	Usable IP Range	Broadcast	Gateway
Students	600	255.255.252.0	/22	10.4.0.0	10.4.0.1 – 10.4.3.254	10.4.3.255	10.4.0.1
Faculty	50	255.255.255.192	/26	10.4.4.0	10.4.4.1 – 10.4.4.62	10.4.4.63	10.4.4.1
Admin	30	255.255.255.192	/26	10.4.4.64	10.4.4.65 – 10.4.4.126	10.4.4.127	10.4.4.65
Servers	4	255.255.255.248	/29	10.4.4.128	10.4.4.129 – 10.4.4.134	10.4.4.135	10.4.4.129
Wi-Fi	20	255.255.255.224	/27	10.4.4.160	10.4.4.161 – 10.4.4.190	10.4.4.191	10.4.4.161

Campus E:

10.5.0.0/16

Department	Hosts	Subnet Mask	CIDR	Network Address	Usable IP Range	Broadcast	Gateway
Students	600	255.255.252.0	/22	10.5.0.0	10.5.0.1 – 10.5.3.254	10.5.3.255	10.5.0.1
Faculty	60	255.255.255.192	/26	10.5.4.0	10.5.4.1 – 10.5.4.62	10.5.4.63	10.5.4.1
Admin	25	255.255.255.224	/27	10.5.4.64	10.5.4.65 – 10.5.4.94	10.5.4.95	10.5.4.65
Servers	4	255.255.255.248	/29	10.5.4.96	10.5.4.97 – 10.5.4.102	10.5.4.103	10.5.4.97
Wi-Fi	20	255.255.255.224	/27	10.5.4.128	10.5.4.129 – 10.5.4.158	10.5.4.159	10.5.4.129

Campus F:

10.6.0.0/16

Department	Hosts	Subnet Mask	CIDR	Network Address	Usable IP Range	Broadcast	Gateway
Students	500	255.255.254.0	/23	10.6.0.0	10.6.0.1 – 10.6.1.254	10.6.1.255	10.6.0.1
Faculty	60	255.255.255.192	/26	10.6.4.0	10.6.4.1 – 10.6.4.62	10.6.4.63	10.6.4.1
Admin	20	255.255.255.224	/27	10.6.4.64	10.6.4.65 – 10.6.4.94	10.6.4.95	10.6.4.65
Servers	4	255.255.255.248	/29	10.6.4.96	10.6.4.97 – 10.6.4.102	10.6.4.103	10.6.4.97
Wi-Fi	20	255.255.255.224	/27	10.6.4.128	10.6.4.129 – 10.6.4.158	10.6.4.159	10.6.4.129

6. Wireless Network Design

WLC Deployment

- WLC placed in **Campus A VLAN 50**
- Manages all APs across six campuses
- AP mode: Lightweight (CAPWAP)

SSID Design

SSID	VLAN	Security
Uni-Student	30	WPA2-Enterprise
Uni-Faculty	20	WPA2-Enterprise
Uni-Guest	50	WPA2-PSK (separate DHCP pool)

7. Routing Design

6.1 Interior Routing (Inside Each Campus): OSPF Multi-Area

OSPF Areas:

Campus	Area ID
A	1
B	2
C	3
D	4
E	5
F	6

- Each campus runs **OSPF internally**.
- Area 0 runs on the **Core Router** at Campus A.
- All campus routers connect to Area 0 using **point-to-point WAN links**.

6.2 Exterior Routing (Between Campuses): EIGRP

- Autonomous System Number: **AS 100**
- Used only between **campus routers** to exchange inter-campus routes.
- Redistribution between OSPF and EIGRP occurs at **Campus A Core Router**.

8. WAN Design

Campus Interconnections:

Campus A ↔ B

Campus A ↔ C

Campus A ↔ D

Campus A ↔ E

Campus A ↔ F

- Backbone routers use /30 links:
    - 10.255.x.x/30

Example:

A–B Link → 10.255.1.0/30

A–C Link → 10.255.2.0/30

...

9. DHCP Design

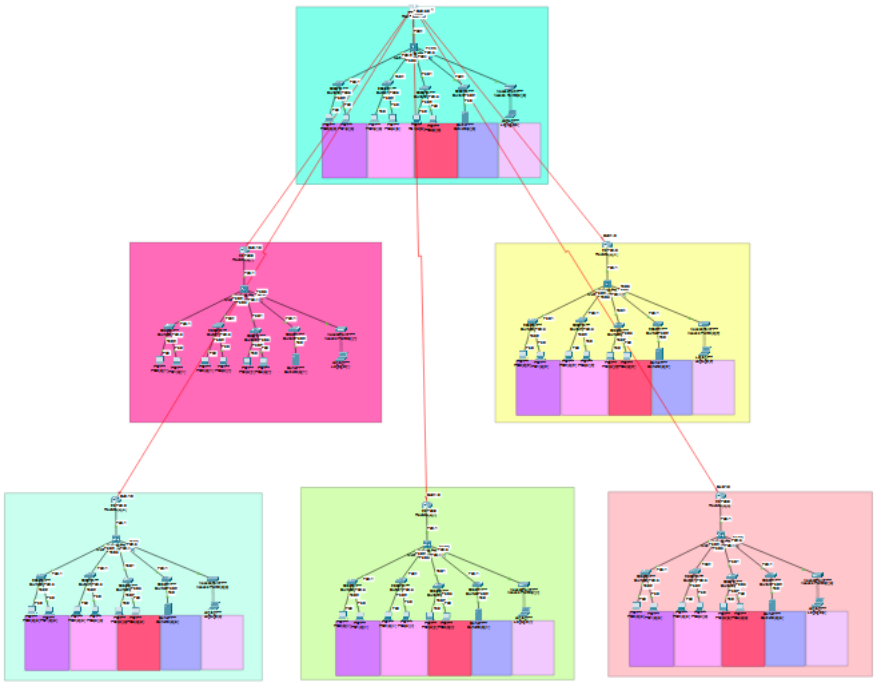
DHCP Servers exist in Campus A (Centralized)

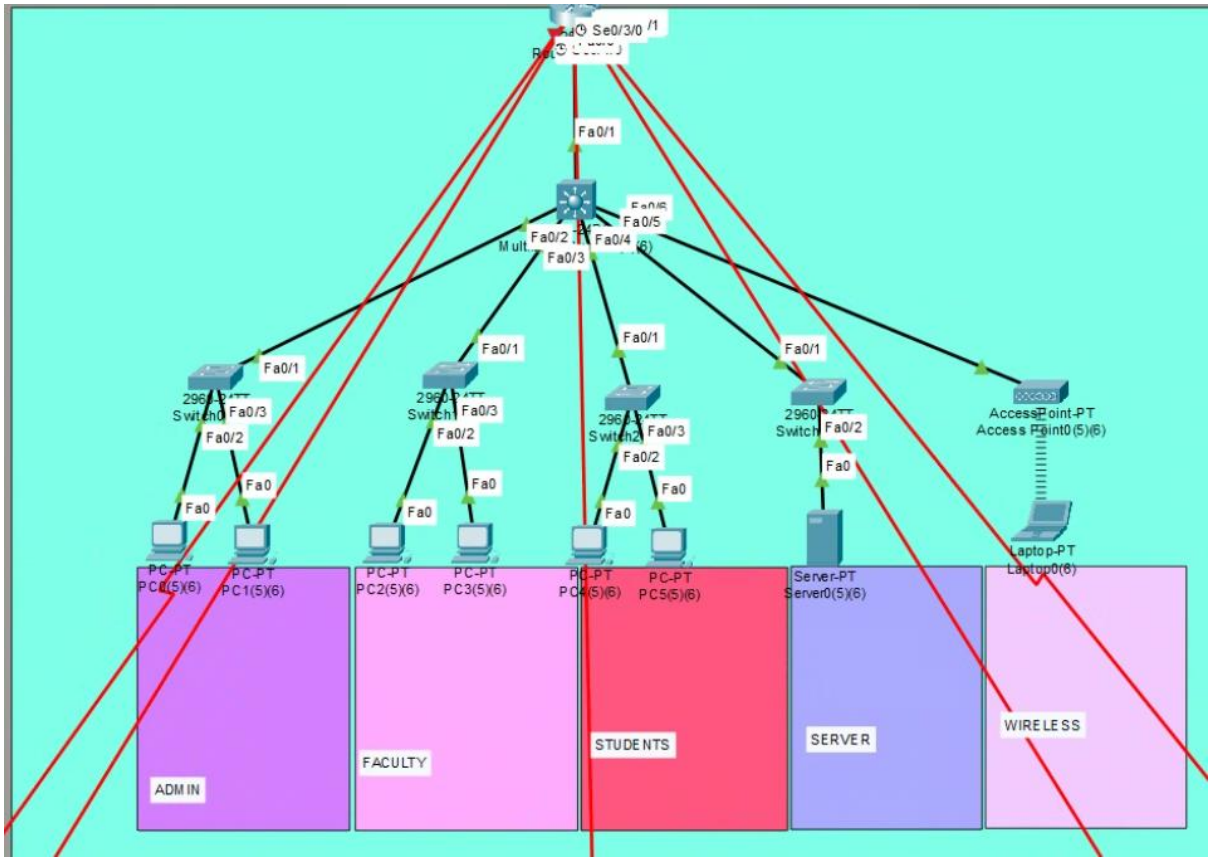
- Router-on-a-Stick subinterfaces forward DHCP requests via **ip helper-address**.
- DHCP Pools for:
  - Admin
  - Student
  - Faculty
  - Servers
  - Wireless
  - Guest WiFi

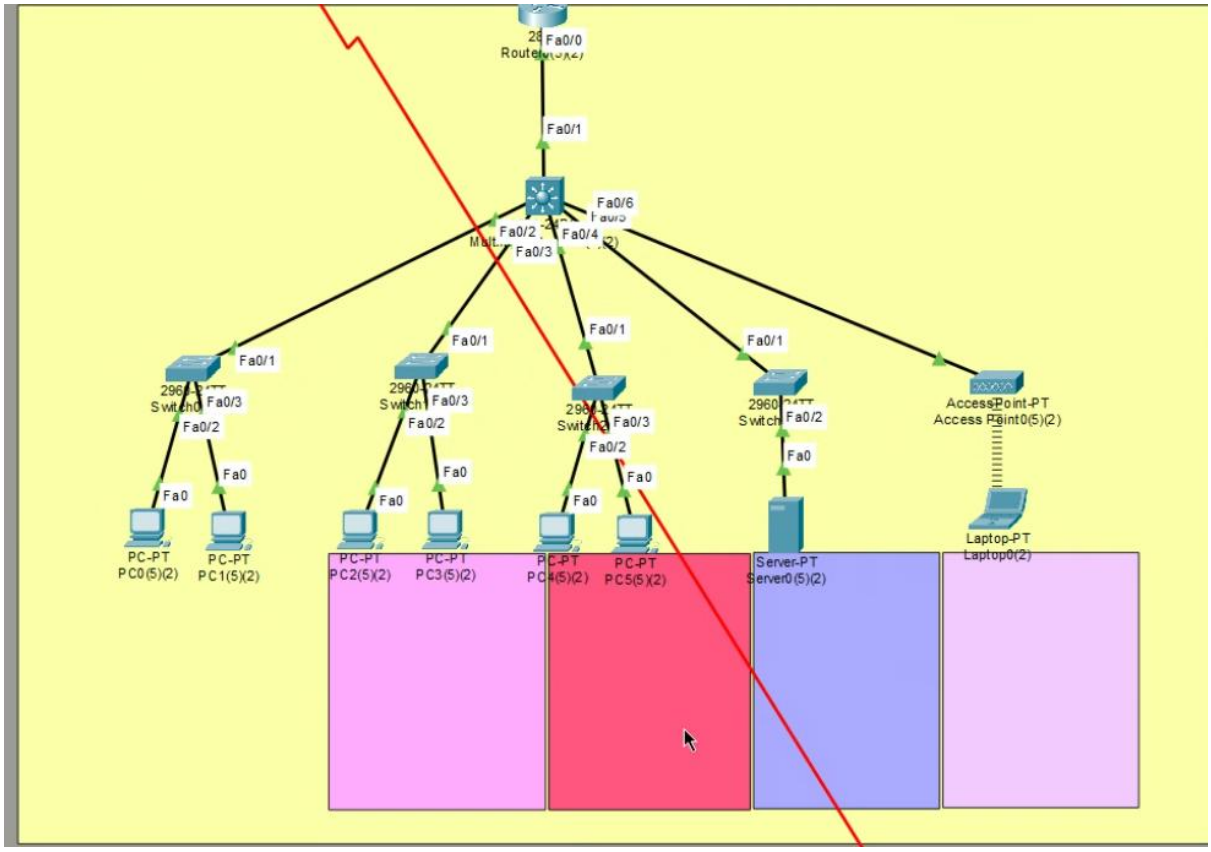
10. Security Measures

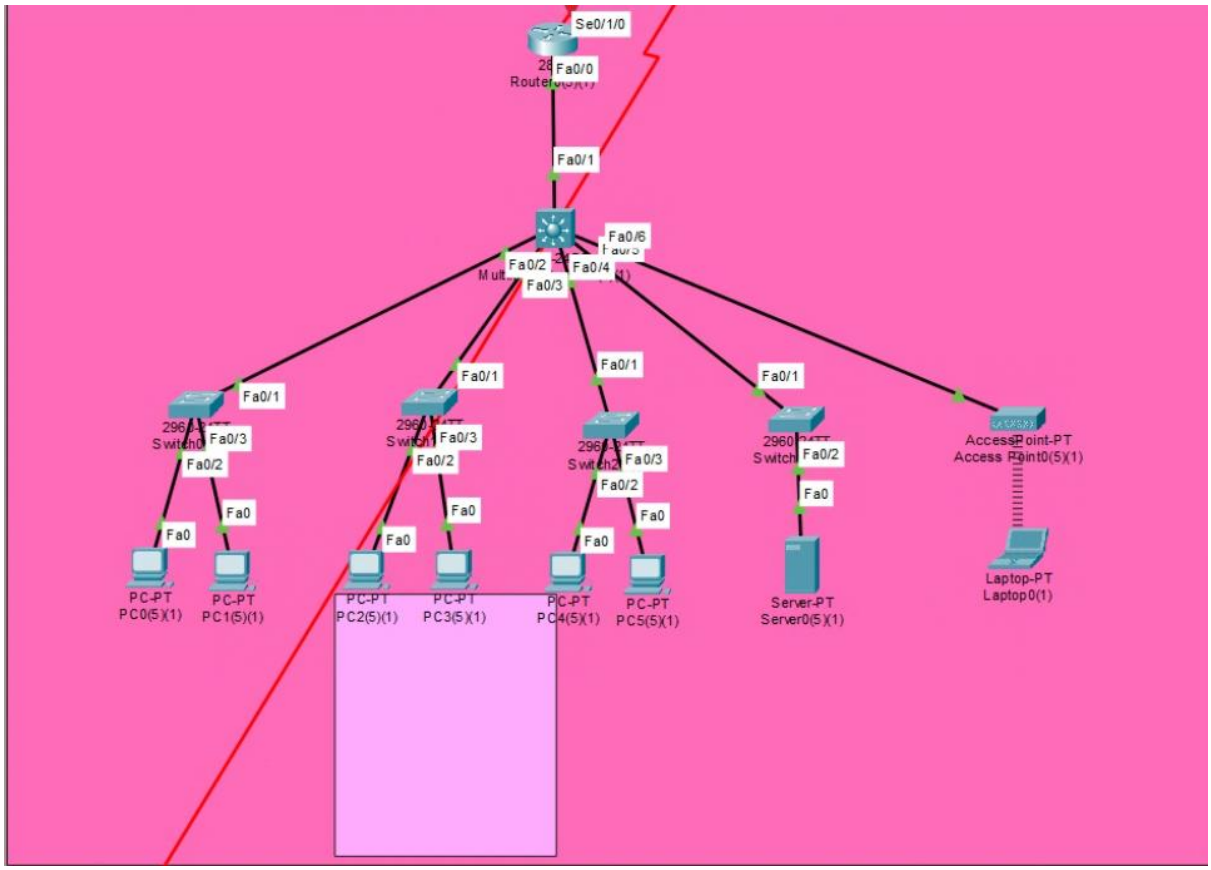
Layer	Security Features
Layer 2	Port security, BPDU Guard, Disabled DTP
Layer 3	ACLs, OSPF Authentication, EIGRP Authentication
Perimeter	NAT overload for Internet
Wireless	WPA2 Enterprise, Guest VLAN isolation
Management	SSH only, SNMPv3

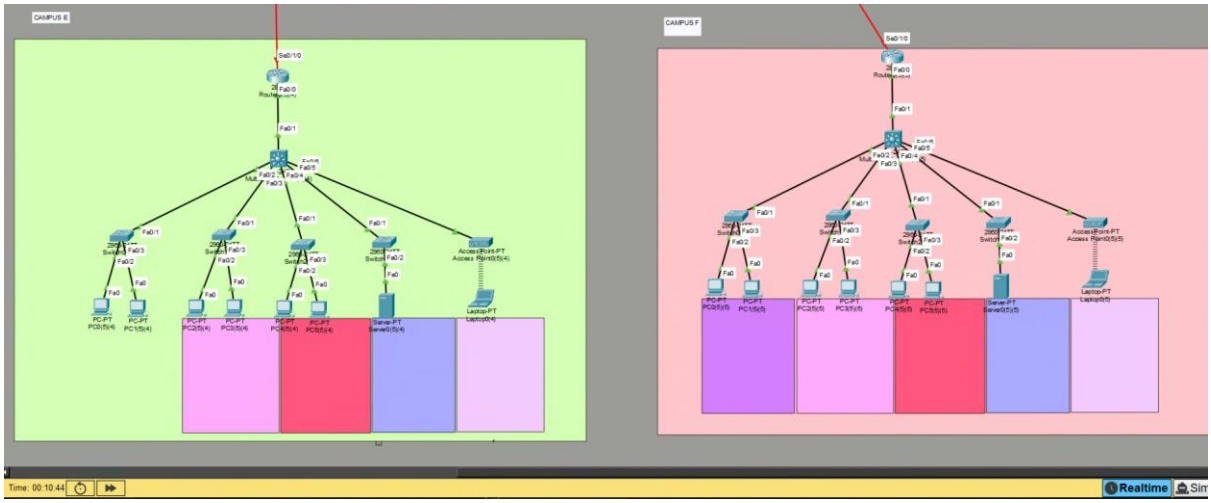
11. Complete Topology Diagram

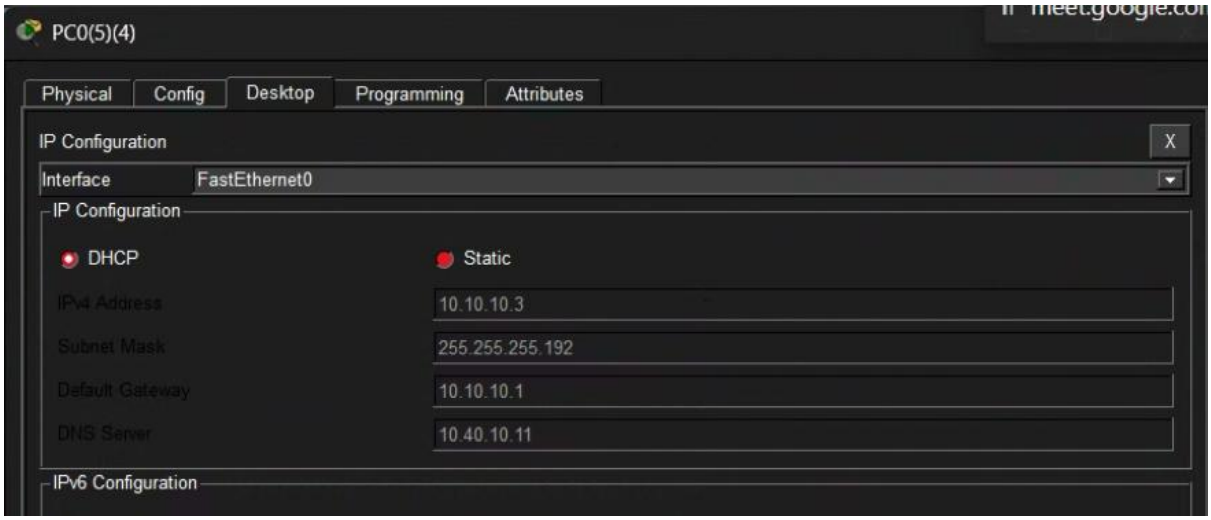


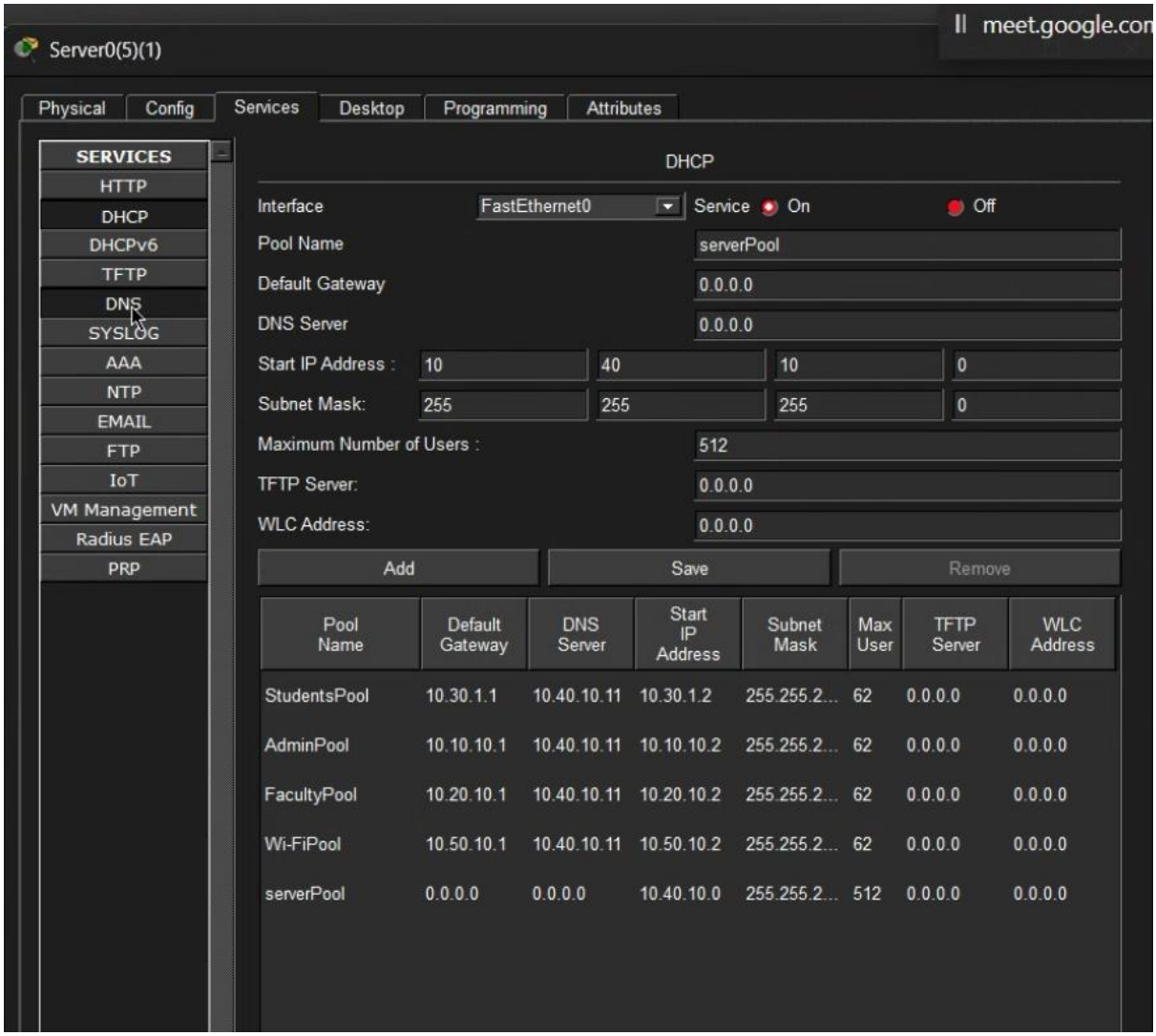


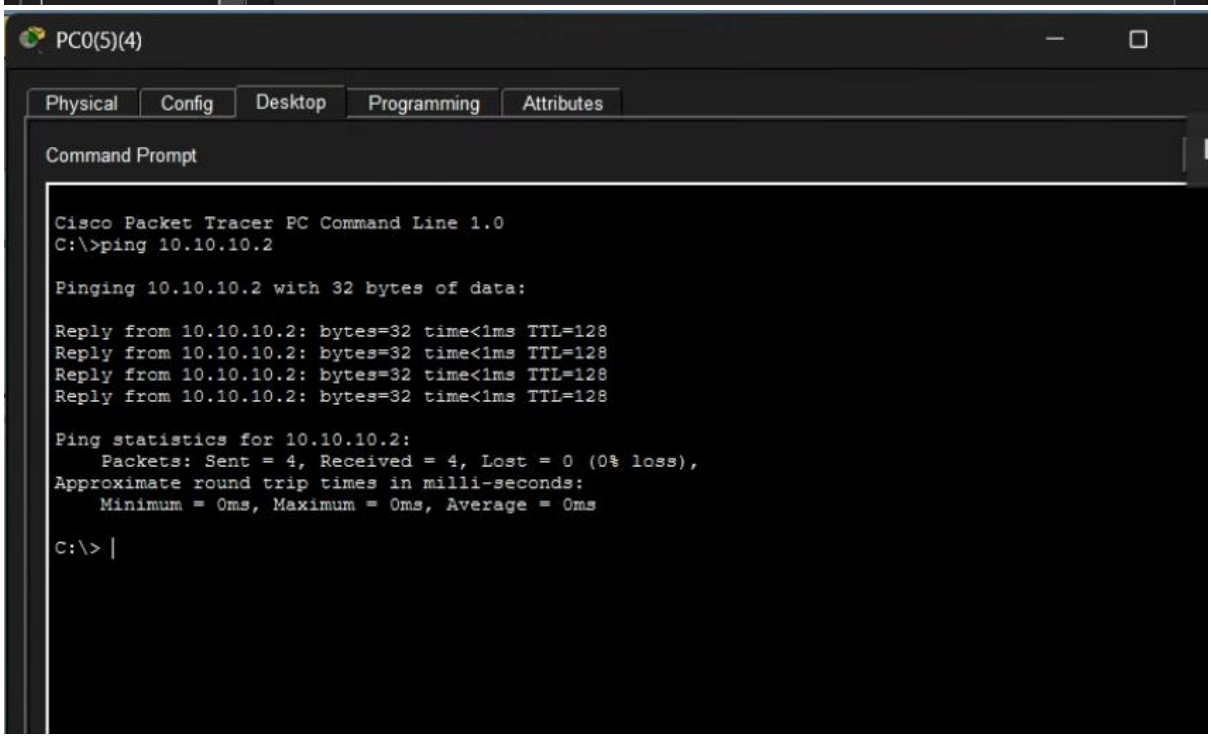

Fa0/0
Fa0/1
Fa0/6
Fa0/2
Fa0/4
Fa0/3
Fa0/1
Fa0/3
Fa0/2
Fa0
PC-PT
PC0(5)(2)
PC-PT
PC1(5)(2)
PC-PT
PC2(5)(2)
PC-PT
PC3(5)(2)
PC-PT
PC4(5)(2)
PC-PT
PC5(5)(2)
Server-PT
Server0(5)(2)
Access Point-PT
Laptop-PT
Laptop0(2)


Se0/1/0
Fa0/0
Fa0/1
Fa0/6
Fa0/2
Fa0/4
Fa0/3
Fa0/1
Fa0/3
Fa0/2
Fa0
PC-PT
PC0(5)(1)
PC-PT
PC1(5)(1)
PC-PT
PC2(5)(1)
PC-PT
PC3(5)(1)
PC-PT
PC4(5)(1)
PC-PT
PC5(5)(1)
Server-PT
Server0(5)(1)
Access Point-PT
Laptop-PT
Laptop0(1)


CAMPUS E
CAMPUS F
Realtime


PC0(5)(4)
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.10.10.3
Subnet Mask
255.255.255.192
Default Gateway
10.10.10.1
DNS Server
10.40.10.11
IPv6 Configuration


meet.google.com
Server0(5)(1)
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
Start IP Address :
10
40
10
0
Subnet Mask:
255
255
255
0
Maximum Number of Users :
512
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
StudentsPool 10.30.1.1 10.40.10.11 10.30.1.2 255.255.2... 62 0.0.0.0 0.0.0.0
AdminPool 10.10.10.1 10.40.10.11 10.10.10.2 255.255.2... 62 0.0.0.0 0.0.0.0
FacultyPool 10.20.10.1 10.40.10.11 10.20.10.2 255.255.2... 62 0.0.0.0 0.0.0.0
Wi-FiPool 10.50.10.1 10.40.10.11 10.50.10.2 255.255.2... 62 0.0.0.0 0.0.0.0
serverPool 0.0.0.0 0.0.0.0 10.40.10.0 255.255.2... 512 0.0.0.0 0.0.0.0


PC0(5)(4)
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.2
Pinging 10.10.10.2 with 32 bytes of data:
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Ping statistics for 10.10.10.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>

12. Packets Transfer

Realtime Simulation

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0(...	PC0(5)(6)	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0(...	Laptop0(6)	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC0(...	PC0(5)(6)	ICMP		0.000	N	2	(edit)	(delete)

PC0(5)(6)

Physical Config Desktop Programming Attributes

Web Browser

< > URL http://10.40.10.10 Go Stop

Cisco Packet Tracer

Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

Quick Links:
A small page
Copyrights
Image page
Image

Welcome to Campus 1 Web Server!

Access Point0(5)(6)

Physical Config Attributes

GLOBAL
Settings
INTERFACE
Port 0
Port 1

Port 1

Port Status: On

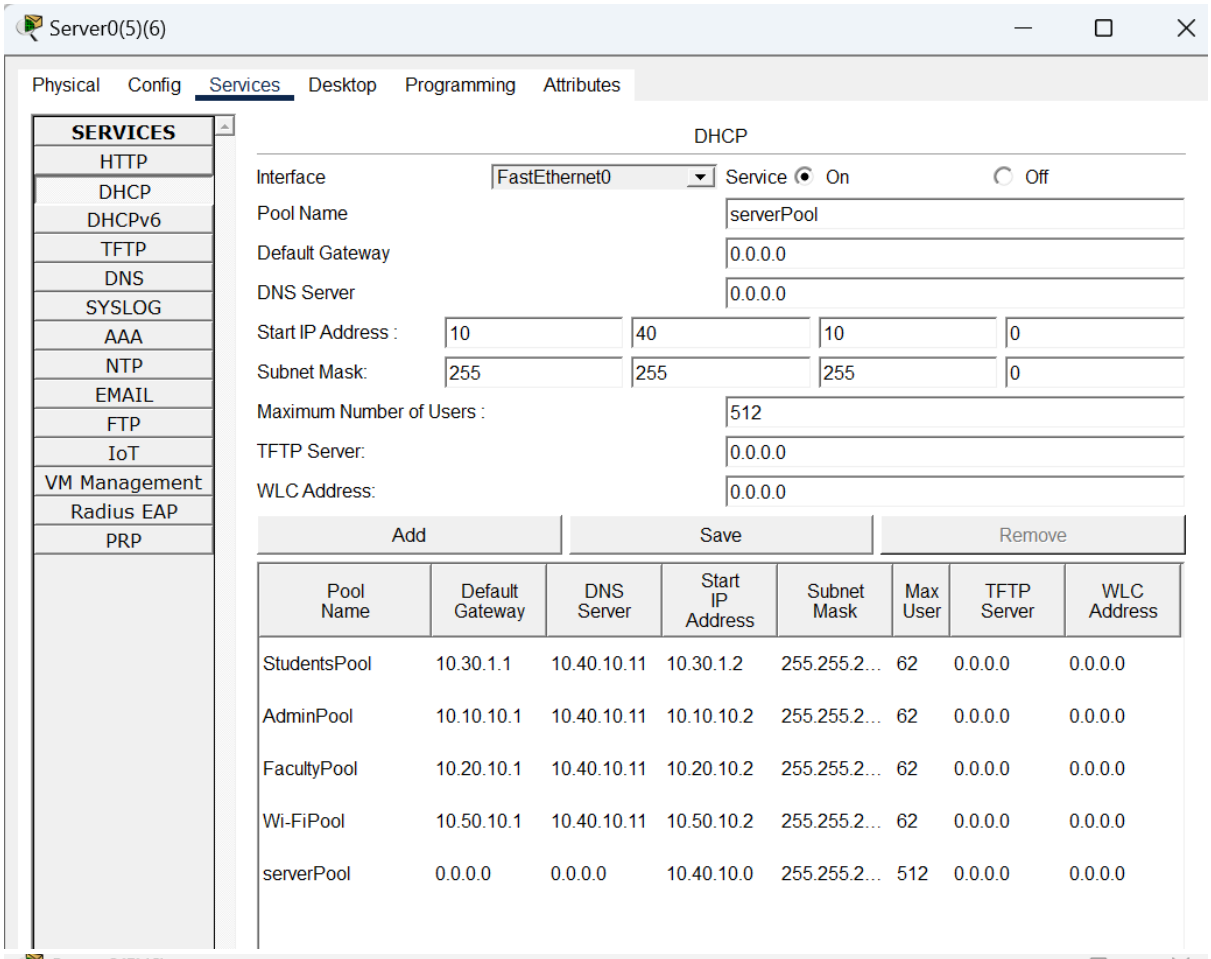
SSID: CampusA_WiFi
2.4 GHz Channel: 1

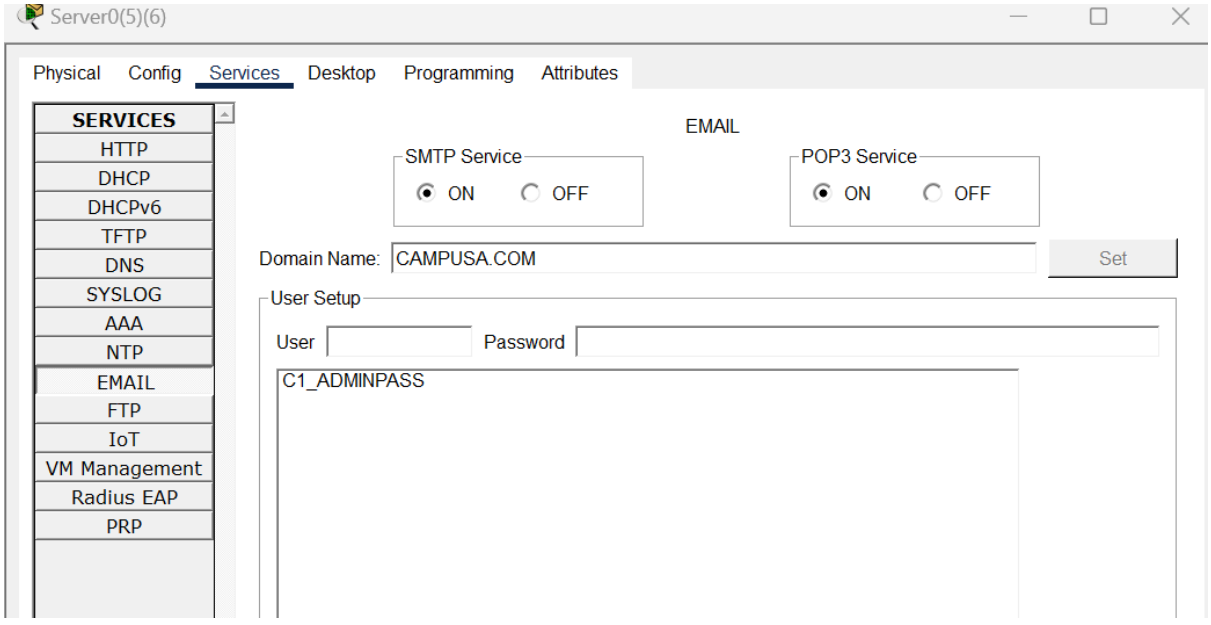
Coverage Range (meters): 140.00

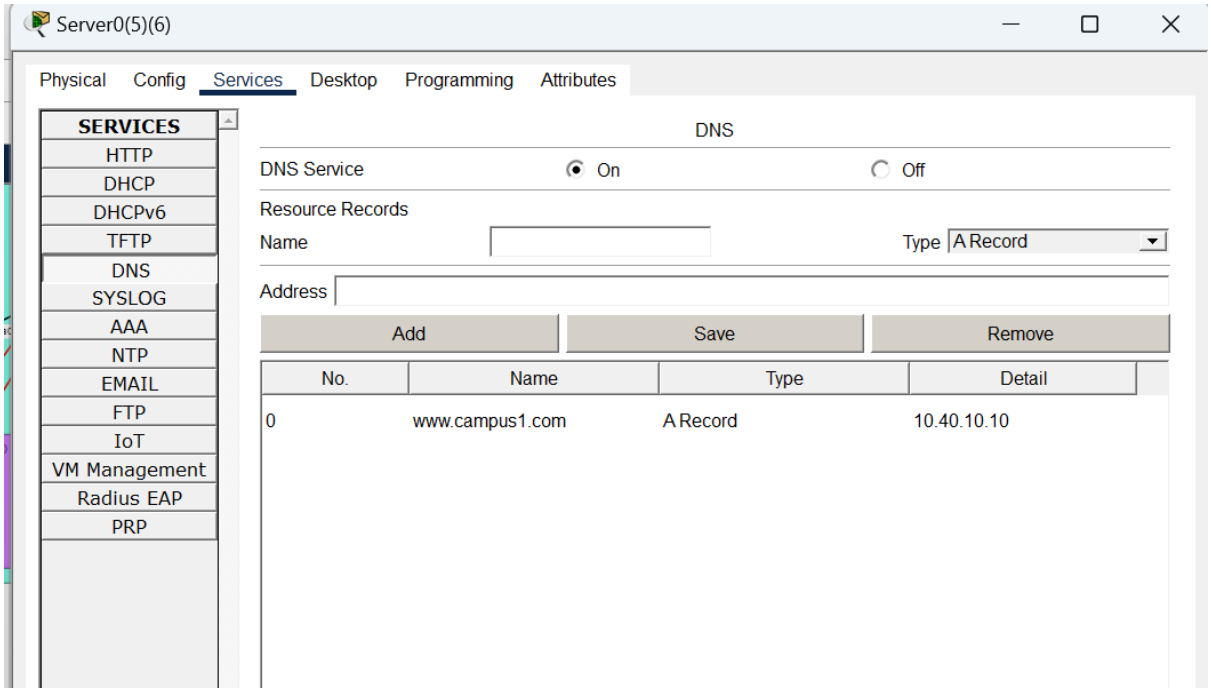
Authentication
Disabled WEP
WPA-PSK WPA2-PSK

WEP Key:
PSK Pass Phrase: 12345678
User ID:
Password:

Encryption Type: AES


Server0(5)(6)
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
Start IP Address :
10
40
10
0
Subnet Mask:
255
255
255
0
Maximum Number of Users :
512
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
StudentsPool
10.30.1.1
10.40.10.11
10.30.1.2
255.255.2...
62
0.0.0.0
0.0.0.0
AdminPool
10.10.10.1
10.40.10.11
10.10.10.2
255.255.2...
62
0.0.0.0
0.0.0.0
FacultyPool
10.20.10.1
10.40.10.11
10.20.10.2
255.255.2...
62
0.0.0.0
0.0.0.0
Wi-FiPool
10.50.10.1
10.40.10.11
10.50.10.2
255.255.2...
62
0.0.0.0
0.0.0.0
serverPool
0.0.0.0
0.0.0.0
10.40.10.0
255.255.2...
512
0.0.0.0
0.0.0.0


Server0(5)(6)
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
EMAIL
SMTP Service
ON
OFF
POP3 Service
ON
OFF
Domain Name:
CAMPUSA.COM
Set
User Setup
User
Password
C1_ADMINPASS



13. ACCESS PORT CONFIGURATION

Switch1: Admin (VLAN 10)

enable
configure terminal

interface fa0/1
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast

interface fa0/2
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast

end
write memory

Switch2: Faculty (VLAN 20)

enable
configure terminal

```
interface fa0/1
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast

interface fa0/2
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast

end
write memory
```

Switch3: Students (VLAN 30)

```
enable
configure terminal

interface fa0/1
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast

interface fa0/2
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast

end
write memory
```

Servers (VLAN 40)

```
enable
configure terminal

interface range fa0/1 - 4
 switchport mode access
 switchport access vlan 40
 spanning-tree portfast

end
write memory
```

6. WIFI ACCESS POINT PORT (MLS Fa0/6)

```
enable
configure terminal

interface fa0/6
 switchport mode access
 switchport access vlan 200
 spanning-tree portfast
 no shutdown

end
write memory
```

DHCP CONFIGURATION (MLS)

```
enable
configure terminal

ip dhcp excluded-address 10.1.0.1 10.1.0.10
ip dhcp excluded-address 10.1.4.1 10.1.4.5
ip dhcp excluded-address 10.1.4.129 10.1.4.135
ip dhcp excluded-address 10.1.4.193 10.1.4.198
ip dhcp excluded-address 10.1.4.225 10.1.4.230

ip dhcp pool STUDENTS
 network 10.1.0.0 255.255.252.0
 default-router 10.1.0.1
 dns-server 10.1.4.195

ip dhcp pool FACULTY
 network 10.1.4.0 255.255.255.128
 default-router 10.1.4.1
 dns-server 10.1.4.195

ip dhcp pool ADMIN
 network 10.1.4.128 255.255.255.192
 default-router 10.1.4.129
 dns-server 10.1.4.195

ip dhcp pool SERVERS
 network 10.1.4.192 255.255.255.248
 default-router 10.1.4.193
 dns-server 10.1.4.195
```

ip dhcp pool WIFI
 network 10.1.4.224 255.255.255.224
 default-router 10.1.4.225
 dns-server 10.1.4.195
end
write memory

14. CLI Configuration (Appendix)

```
 L3SW-1(config)# interface Vlan10
 L3SW-1(config-if)# ip address 10.1.10.1 255.255.255.0
 L3SW-1(config-if)# ip helper-address 10.1.40.10
 L3SW-1(config-if)# no shutdown
 L3SW-1(config-if)# exit

 L3SW-1(config)# interface Vlan20
 L3SW-1(config-if)# ip address 10.1.20.1 255.255.255.0
 L3SW-1(config-if)# ip helper-address 10.1.40.10
 L3SW-1(config-if)# no shutdown
 L3SW-1(config-if)# exit
```

```
L3SW-1(config)# interface Vlan30
L3SW-1(config-if)# ip address 10.1.0.1 255.255.254.0
L3SW-1(config-if)# ip helper-address 10.1.40.10
L3SW-1(config-if)# no shutdown
L3SW-1(config-if)# exit

L3SW-1(config)# interface Vlan40
L3SW-1(config-if)# ip address 10.1.40.1 255.255.255.0
L3SW-1(config-if)# no shutdown
L3SW-1(config-if)# exit
```

```
L3SW-1(config)# interface Vlan50
L3SW-1(config-if)# ip address 10.1.50.1 255.255.255.0
L3SW-1(config-if)# ip helper-address 10.1.40.10
L3SW-1(config-if)# no shutdown
L3SW-1(config-if)# exit

// --- Router Link Configuration (L3SW-1 side is .2) ---
L3SW-1(config)# interface Fa0/1
L3SW-1(config-if)# no switchport
L3SW-1(config-if)# ip address 10.255.1.2 255.255.255.0
L3SW-1(config-if)# no shutdown
L3SW-1(config-if)# end
L3SW-1# write memory
```

```
Router-1(config)# interface Fa0/1
Router-1(config-if)# ip address 10.255.1.1 255.255.255.0
Router-1(config-if)# no shutdown
Router-1(config-if)# exit

// --- Serial Link Configuration (To R2) ---
Router-1(config)# interface S0/0/0 // To Router 2
Router-1(config-if)# ip address 192.168.1.1 255.255.255.252
// NOTE: One end of a serial link must have the clock rate command if it's the DCE
// Router-1(config-if)# clock rate 128000
Router-1(config-if)# no shutdown
Router-1(config-if)# exit
```

```
// --- EIGRP Routing Protocol (AS 10) ---
Router-1(config)# router eigrp 10
// Advertise all 10.1.x.x networks and the link to the L3 Switch
Router-1(config-router)# network 10.1.0.0 0.0.255.255
// Advertise the 192.168.1.x serial links
Router-1(config-router)# network 192.168.1.0 0.0.0.255
Router-1(config-router)# no auto-summary
Router-1(config-router)# end
Router-1# write memory
```

```
Router-1(config)# access-list 1 permit 10.1.0.0 0.0.255.255

// Step 2: Implement PAT (NAT Overload) using the outside interface's IP
Router-1(config)# ip nat inside source list 1 interface S0/0/0 overload

// Step 3: Define interface roles
Router-1(config)# interface Fa0/1
Router-1(config-if)# ip nat inside
Router-1(config-if)# exit

Router-1(config)# interface S0/0/0
Router-1(config-if)# ip nat outside
Router-1(config-if)# exit
```

```
Router-1(config)# no router eigrp 10

// Step 2: Enable OSPF process (Process ID 1)
Router-1(config)# router ospf 1
Router-1(config-router)# router-id 1.1.1.1 // Unique ID for RT-1

// Step 3: Advertise all Campus 1 local networks (10.1.x.x) and L3 link
Router-1(config-router)# network 10.1.0.0 0.0.255.255 area 0
Router-1(config-router)# network 10.255.1.0 0.0.0.255 area 0

// Step 4: Advertise the WAN link to RT-2
Router-1(config-router)# network 192.168.1.0 0.0.0.3 area 0
```

```
Router-1# configure terminal
// Configure Console access with a password
Router-1(config)# line console 0
Router-1(config-line)# password c1_console
Router-1(config-line)# login
Router-1(config-line)# exit

// Configure VTY (Telnet/SSH) access with a password
Router-1(config)# line vty 0 4
Router-1(config-line)# password c1_vty
Router-1(config-line)# login
Router-1(config-line)# exit
```

```
Router-1(config)# // ACL to match traffic coming into the campus (Inbound on S0/0/
Router-1(config)# access-list 101 permit ip any any established
Router-1(config)# access-list 101 deny ip any any log // Deny everything else

Router-1(config)# interface S0/0/0
Router-1(config-if)# ip access-group 101 in
Router-1(config-if)# exit

Router-1(config)# end
Router-1# write memory
```

15. Why This Design is the Best

✓ **Scalable**

Using 10.0.0.0/8 with /16 per campus supports future population growth.

✓ **Fast Convergence**

OSPF Multi-Area ensures efficient routing inside each campus.

✓ **Robust Inter-Campus Connectivity**

EIGRP gives high performance and load balancing across campuses.

✓ **Centralized Wireless Control**

WLC in Campus A reduces management overhead.

✓ **High Security**

ACLs + WPA2 Enterprise + OSPF/EIGRP authentication + VLAN segmentation.

✓ **Easy Management**

Campus A hosts all important services (DHCP, WLC, Core Routing).

16. Conclusion

The designed network fulfills all requirements of a large enterprise-level university system. It ensures:

- High scalability
- Centralized authentication
- Secure access control
- High-speed wired and wireless connectivity
- Fast inter-campus communication
- Structured routing using OSPF + EIGRP redistribution
- Clean VLAN separation for departments

This report, along with the full CLI configuration, forms the complete submission for the university network design project.

THE END